

Deliverables	Explanation
UART Read	Be able to properly read the incoming data from the car's ECU.
UART Transmit	Be able to properly write the data from the FPGA to the Car.
UART Read FIFO	This FIFO should be able to store enough of the read data until the FPGA side of the program handshakes data and properly requests it
UART TX FIFO	this FIFO should be able to take data from the fast moving FPGA side and properly provide the data to the slower UART controller
FPGA MC Read Controller	This should be able to take data from the Read FIFO and convert that data into user readable information
FPGA MC TX controller	This should be able to take user input on what code or information needs to be transmitted into the ECU
FPGA RAM FIFO	This should be able to store and handshake data with the VGA controller when there is a data request.
VGA Data Handshake Module	This module is meant for the handshake between the RAM FIFO and the processing modules of the VGA
VGA Frame Loader Module	This module is meant to make and process all of the VGA data and create the graph and other data that we want

VGA Fram Buffer Module

This module loads the data from the FPGA to the monitor

Commitment	Goal
Properly read incoming data with 70% accuracy	Read data with 100% accuracy and be able to diagnose errors if needed and alert system
ECU is able to properly read data string with 70% accuracy	Write data with 100% accuracy and have system alerts for if write goes wrong
FIFO is able to send data to the faster moving FPGA with proper handshake protocol	FIFO is able to send data to faster moving FPGA with proper handshake protocol with memory management in store
FIFO is able to take necessary data and provide it to the TX controller	FIFO is able to take necessary data and provide it to TX controller with memory management
Able to properly dissect information code read from ECU	Able to properly dissect information code read from ECU, utilizes lookup tables and efficient string parsing to maximize programming efficiency
Able to parse user inputted information and create correct transmit code	Able to parse user inputted information and create correct transmit code as well as using efficient memory management techniques and maximize LUT use
Stores memory in DDRam and outputs correct address	Stores memory in DDRam and outputs correct address with efficient storage and load protocols
Able to properly retrieve the data and process it whenever the frame rate is updated	Able to properly retrieve the data and process it whenever the frame rate is updated
Create the correct image vectors to display raw data	Create the correct image vectors to display graph of trip

Be able to display any sort of data	Be able to display graph data within specified frame rate
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Stretch Goal

Read data with 100% accuracy and be able to diagnose errors if needed and alert system and user and be able to make an error detection system

Write data with 100% accuracy and have system alerts and error detection in place in case it happens
FIFO is able to send data to faster moving FPGA with proper handshake protocol and EFFICIENT memory management is in place

FIFO is able to take necessary data and provide it to TX controller with memory management

Able to properly dissect information code read from ECU, utilizes lookup tables and efficient string parsing to maximize programming efficiency as well as proper memory management

Able to parse user inputted information and create correct transmit code as well as using efficient memory management techniques and maximize LUT use

Stores memory in DDRam and outputs correct address with efficient storage and load protocols

Able to properly retrieve the data and process it whenever the frame rate is updated
Create the correct image vectors
display graph of trip as well as average stats over trip

Be able to display graph data as well as
average statistics data within frame
rate and support capabilities for faster
frame rate